

Income, Consumption Smoothing, and Household Portfolio Choices in India

Evidence from CMIE Household Microdata, January–December 2025

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Abstract

This project leverages large-scale microdata from the Centre for Monitoring Indian Economy (CMIE) to analyze the financial behavior, consumption smoothing, and savings anomalies of Indian households. Through an end-to-end data engineering pipeline featuring out-of-core SQL merges (DuckDB), I construct a robust master panel to test core economic theories. I estimate the Marginal Propensity to Consume (MPC) and employ a household fixed-effects specification to test the Permanent Income Hypothesis (PIH) by decomposing within-household income shocks. Furthermore, I investigate the "co-holding puzzle"—the simultaneous holding of high-cost debt and illiquid savings—and find that formal employment's assumed link to savings discipline appears largely driven by mechanical payroll deductions rather than genuine intention-behavior follow-through.

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1 Introduction

1.1 Background and Motivation

Household financial decisions determine how income is converted into current consumption, precautionary saving, long-term asset accumulation, and debt repayment. These choices are particularly consequential in India, where household portfolios have historically been concentrated in real assets and gold, formal retirement saving is limited for many households, and borrowing can occur through both formal and informal channels (Badarinza, Balasubramaniam, and Ramadorai, 2017). Yet high-frequency evidence on how the same households adjust income, consumption, saving intentions, and debt over time remains comparatively scarce.

This study uses CMIE household microdata for the full calendar year 2025 to examine three connected dimensions of household finance. First, it estimates the relationship between income and consumption and evaluates whether households respond differently to persistent and temporary income variation. Second, it studies the apparent puzzle of households simultaneously holding high-cost debt and illiquid savings. Third, it examines whether formal employment predicts the conversion of stated saving intentions into realised saving, distinguishing payroll-linked provident-fund saving from voluntary gold saving.

1.2 Problem Statement and Research Gap

The standard consumption-smoothing framework predicts that households should respond more strongly to persistent income changes than to temporary shocks (Friedman, 1957). At the same time, observed household portfolios can appear inconsistent with simple cost minimisation: a household may retain low-return or illiquid savings while borrowing from expensive sources. This pattern may instead reflect precautionary liquidity needs, transaction constraints, institutional rules, or measurement choices (Telyukova, 2013; Gross and Souleles, 2002).

Existing evidence on Indian household balance sheets documents important portfolio regularities, including substantial real-asset and gold holdings and limited retirement assets (Badarinza, Balasubramaniam, and Ramadorai, 2017). The remaining gap is a unified, recent panel analysis that links monthly income and expenditure to wave-level debt, savings, employment, and stated intentions. This paper addresses that gap with a transparent panel-construction procedure and a set of weighted, household-clustered regressions.

1.3 Research Objectives and Questions

The objectives are to:

1. construct a reproducible household panel from four CMIE datasets covering January–December 2025;
2. estimate the marginal propensity to consume (MPC) from household income and test the permanent income hypothesis (PIH);
3. describe and model the co-holding of high-cost debt and illiquid savings; and
4. evaluate whether formal employment is associated with follow-through on intended provident-fund and gold saving.

The empirical analysis answers the following research questions:

- RQ1: What is the relationship between monthly household income and expenditure in the CMIE sample?
- RQ2: Is the consumption response to permanent income different from the response to transitory income?
- RQ3: Which characteristics are associated with holding high-cost debt and illiquid savings at the same time?
- RQ4: Does formal employment predict a smaller intention–behaviour gap for provident-fund and gold saving?

2 Literature Review

2.1 Consumption Smoothing and Permanent Income

The permanent income hypothesis argues that consumption should principally reflect expected long-run resources rather than short-lived income movements (Friedman, 1957). In an empirical panel, this implication can be investigated by separating each household's income into a household-specific mean and a deviation from that mean. A larger consumption coefficient for the former than for the latter is consistent with consumption smoothing. Evidence that consumers react to liquidity changes, however, also shows that credit constraints and precautionary motives can generate departures from the simplest PIH prediction (Gross and Souleles, 2002).

2.2 The Co-Holding Puzzle

The co-holding puzzle concerns households that carry costly debt while retaining savings that, on their face, could be used to reduce that debt. Rather than treating this pattern as automatically irrational, the literature emphasises liquidity needs: savings may insure cash-only expenditures or other states in which consumer credit cannot be used (Telyukova, 2013). For India, the distinction is especially important because household assets, debt sources, and access to formal financial instruments vary widely across regions, demographic groups, and employment arrangements (Badarinza, Balasubramaniam, and Ramadorai, 2017).

2.3 Formal Employment and Saving Behaviour

Formal employment may influence household saving through regular earnings, payroll-linked retirement contributions, and access to financial institutions. An observed association between formal employment and provident-fund ownership nevertheless need not imply a broad behavioural advantage: it may arise mechanically from the institutional design of payroll deductions. Comparing a payroll-linked saving vehicle (provident fund) with an active-choice vehicle (gold) provides a useful diagnostic. A formal-employment effect confined to provident-fund follow-through would be consistent with an institutional mechanism rather than a general improvement in financial discipline.

2.4 Hypotheses

- H1.** The consumption response to permanent income is greater than the response to transitory income.
- H2.** Income volatility and formal employment are associated with the co-holding indicator after conditioning on demographics and location.
- H3.** Formal employment is associated with a smaller intention–behaviour gap for provident-fund saving.
- H4.** If the formal-employment effect reflects a general behavioural advantage, it is also associated with a smaller gold-saving gap.

3 Data Preparation

3.1 Data Source, Period, and Unit of Analysis

The study uses CMIE household microdata for January through December 2025 (Centre for Monitoring Indian Economy, 2025). Four modules are combined:

- (a) *Household Income*, providing monthly total household income (TOT_INC);
- (b) *Consumption Pyramids*, providing monthly total expenditure (TOT_EXP);
- (c) *People of India*, providing member-level demographic and employment information; and
- (d) *Aspirations of India*, providing wave-level saving, borrowing, and intention variables.

The monthly backbone is the household-month. The People of India and Aspirations of India modules are wave-based and are therefore represented at the household-wave level, with waves defined as January–April, May–August, and September–December. The final analysis deliberately uses an unbalanced panel in order not to exclude households that are observed in fewer than all three waves. In the executed panel, 89.07% of households appear in all three waves, 4.05% appear in two waves, and 6.88% appear in one wave.

3.2 Cleaning and Harmonisation Pipeline

The data preparation follows the eight-stage pipeline implemented in `Indian_Household_Data_Cleaning.ipynb` ¹:

Stage 1: **Schema standardisation.** Column names are stripped and converted to upper case. All monthly files are scanned before processing so that missing columns can be added and inconsistent data types can be resolved conservatively.

Stage 2: **Non-response and household-change flags.** Observations are retained and explicitly flagged as responding or non-responding. A family-shift flag identifies a potential change in the family associated with a household identifier.

Stage 3: **Time-granularity alignment.** Income and consumption data are retained at the monthly level. Repeated copies of wave-based modules are collapsed to

¹Source code available at: <https://github.com/Ravishek-Projects/Behavioral-Household-Finance-in-India>

a single household-wave observation using the last non-missing value within the wave.

Stage 4: **Key audit.** The pipeline checks that the intended primary keys are unique before merging, preventing join fan-out from silently multiplying observations.

Stage 5: **Variable coding and outlier treatment.** Yes/no fields are recoded as Boolean variables. Negative monetary values are set to missing, and monetary variables are winsorised at the 99th percentile within the specified wave and occupation groups.

Stage 6: **Household-level aggregation.** Member-level data are aggregated to the household-wave level. Head-of-household characteristics, counts of members/adults/employed persons, a formal-employment indicator, and household asset flags are constructed.

Stage 7: **Derived indicators.** High-cost debt, illiquid savings, co-holding, income volatility, and intention-behaviour gaps are constructed as described below.

Stage 8: **Out-of-core merge.** DuckDB joins the cleaned tables and writes the master panel to Parquet, avoiding a memory-intensive in-memory merge.

The data-generation and preprocessing procedures are designed to be sufficiently transparent that another researcher could reconstruct the dataset. All primary-key audits pass after the time-granularity alignment, and the exact CMIE extract dates and licensed-variable definitions are recorded.

3.3 Constructed Variables

Table 1 defines the main variables. The high-cost debt indicator is true when a household reports borrowing from at least one of: lender, credit card, non-banking financial company, microfinance institution, or chit fund. Illiquid savings is true when the household reports savings in a provident fund, life insurance, pension scheme, or fixed deposit. The co-holding indicator requires both conditions to hold.

Table 1: Main analysis variables

Variable	Definition
TOT_INC	Total monthly household income.
TOT_EXP	Total household expenditure in the reporting frequency supplied by the CMIE consumption module.
INC_CV	Coefficient of variation of TOT_INC over 2025; calculated only for households with at least eight non-missing income months.
HIGH_COST_DEBT_FLAG	Indicator for at least one reported high-cost borrowing source.
ILLIQUID_SAVINGS_FLAG	Indicator for at least one reported illiquid saving vehicle.
COHOLD_FLAG	Indicator equal to one when both high-cost debt and illiquid savings indicators equal one.
HAS_FORMAL_EMPLOYED_MEM	Household has at least one employed member classified as formal under the project rule.
GAP_PF	Indicator of non-realisation of a stated provident-fund saving intention between the January–April and May–August waves.
GAP_GOLD	Indicator of non-realisation of a stated gold-saving intention between the January–April and May–August waves.

Note: The formal-employment rule requires employment, full-time status, and an occupation not classified as informal by the project pipeline. Exact CMIE questionnaire wording should be checked against the licensed data documentation before external publication.

3.4 Descriptive Statistics

Table 2 summarises key checks from the executed analysis. Asset rates use the wave-level household weights. The near-universal real-estate (99.09%) and gold (99.67%) participation rates warrant close scrutiny. Two unrelated asset classes both approaching 100% strongly suggest a survey coding or default-imputation issue in the CMIE data (e.g., misinterpreting ancestral ownership as active household title) rather than true individual ownership rates, and these variables must be used cautiously in downstream analysis.

Table 2: Panel composition and selected descriptive checks

Item	Value
Household-month observations	1,876,151
Unique households	170,867
Unique household-wave observations	482,159
Households observed in all three waves	89.07%
Mean monthly income (unweighted)	INR 40,256.10
Mean expenditure (unweighted)	INR 23,251.69
Mean expenditure / mean income	0.578
Weighted real-estate saving participation	99.09%
Weighted gold saving participation	99.67%
Weighted provident-fund saving participation	15.29%
Weighted mutual-fund saving participation	3.19%
Weighted high-cost debt participation	6.45%

Note: The panel counts and descriptive statistics are reproduced from the executed project analysis. Weighted participation rates use R_HH_WGT_W; income-consumption regressions use R_HH_WGT_MS_INC.

4 Research Methodology

4.1 Research Design and Inference

The analysis is observational and combines household-month and household-wave regressions. Models are estimated by weighted least squares (WLS), using CMIE’s monthly income weight for income–expenditure specifications and the wave household weight for wave-level specifications. Standard errors are clustered by household identifier to account for repeated observations of the same household. Coefficients therefore describe conditional associations, not causal effects.

The methodology includes:

- **Research design**— observational panel combining cross-sectional and time-series dimensions.
- **Theoretical framework**— permanent income hypothesis and co-holding liquidity theory.

- **Estimation**— weighted least squares with household-clustered standard errors.
- **Model evaluation**— R^2 , Wald tests, and robustness specifications.
- **Robustness tests**— income trimming, sample restrictions, and alternative dependent variables.

4.2 Income–Consumption Model

The baseline consumption model is:

$$C_{it} = \alpha + \beta Y_{it} + \varepsilon_{it}, \quad (1)$$

where C_{it} is total expenditure and Y_{it} is total monthly income for household i in month t . The coefficient β is interpreted as the estimated MPC. A robustness estimate excludes the upper one percent of the non-missing income distribution. This trimming changes the estimation sample only; it does not alter the raw data stored in the master panel.

4.3 Household Fixed-Effects Decomposition and the PIH

Identification Strategy: By separating income into its household-specific mean ($\bar{Y}_i$) and within-household deviation (Y_{it}^T), this specification is equivalent to a household fixed-effects (within) estimator, which absorbs time-invariant unobserved household heterogeneity. It does not, however, address time-varying confounders—for instance, a shock (job loss, illness) that moves income and consumption in the same period—so the within-household coefficient (β_T) should still be read as a conditional association rather than a causal effect.

For households with at least eight non-missing monthly income observations, income is separated into a household-specific mean and a deviation from that mean:

$$\bar{Y}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} Y_{it}, \quad (2)$$

$$Y_{it}^T = Y_{it} - \bar{Y}_i. \quad (3)$$

The PIH specification is:

$$C_{it} = \alpha + \beta_P \bar{Y}_i + \beta_T Y_{it}^T + u_{it}. \quad (4)$$

The hypothesis $H_0 : \beta_P = \beta_T$ is tested with a Wald test. A larger β_P is consistent with stronger consumption adjustment to persistent income than to temporary fluctuations.

4.4 Co-Holding Model

At the household-wave level, the co-holding linear probability model is:

$$\text{CoHold}_{iw} = \alpha + \gamma_1 \text{IncCV}_i + \gamma_2 \text{Formal}_{iw} + \mathbf{X}'_{iw} \boldsymbol{\delta} + \lambda_s + \eta_e + v_{iw}, \quad (5)$$

where $\mathbf{X}_{iw}$ includes head-of-household age and household size; λ_s and η_e are state and education-group fixed effects. Two checks probe interpretation: (i) the model is re-estimated without `INC_CV`, which avoids excluding households without sufficient income history; and (ii) the dependent variable is changed to high-cost debt within the subset already holding provident-fund savings. The latter check asks whether the apparent formal-employment association is mechanically generated by payroll-linked provident-fund ownership.

4.5 Intention–Behaviour Gap Model

Among households that stated an intention to save in a particular vehicle, the following model is estimated:

$$\text{Gap}_{iw} = \alpha + \theta_1 \text{IncCV}_i + \theta_2 \text{Formal}_{iw} + \mathbf{X}'_{iw} \boldsymbol{\delta} + \lambda_s + \eta_e + e_{iw}. \quad (6)$$

The model is estimated separately for provident-fund and gold saving. The comparison distinguishes an automatic, payroll-linked saving channel from an active discretionary saving choice.

5 Results

5.1 Income, Expenditure, and Consumption Smoothing

The full-sample WLS model estimates an MPC of 0.209 ($p < 0.001$). After removing the upper one percent of non-missing income observations, the estimated MPC rises to 0.261 ($p < 0.001$). This difference indicates sensitivity to the upper tail, although both estimates remain positive and precisely estimated. Because this estimate falls below the conventional benchmark for MPC, the result is reported as a descriptive estimate pending final validation of the expenditure variable's construction.

The permanent-income model provides stronger evidence of consumption smoothing. The estimated response to permanent income is 0.320, compared with 0.053 for transitory income. The Wald test rejects equality ($F = 19,642.53$, $p < 0.001$). Thus, within this

specification, household consumption is considerably more responsive to the household-specific average income than to temporary deviations around that average.

The overall expenditure-to-income ratio of 0.578 is unusually low for this population, implying an unrealistic savings rate of over 40%. This is a known structural artifact where broad surveys heavily underreport consumption compared to detailed diaries (Deaton, 1997). Consequently, the MPC estimates reported in Table 3 should be interpreted strictly as lower bounds, and not as final point estimates, until the exact construction of TOT_EXP is resolved against the CMIE data dictionary.

Table 3: Income-consumption and permanent-income estimates

Specification / coefficient	Estimate	Clustered SE	Observations	Additional result
Baseline MPC: TOT_INC	0.2085	0.0014	1,342,019	$R^2 = 0.335$
Trimmed MPC: TOT_INC	0.2610	0.0013	1,328,604	$R^2 = 0.370$
PIH: permanent income	0.3199	0.0018	1,181,635	$R^2 = 0.474$
PIH: transitory income	0.0529	0.0008	1,181,635	$R^2 = 0.474$
Wald test: permanent = transitory $F = 19,642.53, p < 0.001$				

Note: All models are WLS with standard errors clustered at the household level. The PIH estimation uses households with at least eight observed income months. The trimmed model removes the upper one percent of the *non-missing* income distribution; missing-income observations are excluded by the model's complete-case requirement.

5.2 Co-Holding of High-Cost Debt and Illiquid Savings

The weighted co-holding rate is 5.96% in the wave-level sample. In the primary co-holding model, the coefficient on income volatility is small and statistically insignificant (-0.0003 , $p = 0.862$). Formal employment has a positive association of 1.61 percentage points ($p < 0.001$), conditional on age, household size, state, and education group. The formal-employment coefficient remains positive (1.43 percentage points, $p < 0.001$) when the model is estimated without INC_CV in the larger sample.

This association should not be read as proof that formal workers have a stronger underlying tendency to borrow while saving. When the analysis is restricted to households that already have provident-fund savings, the formal-employment coefficient in a high-cost-debt model is 0.0003 ($p = 0.947$). The disappearance of the association is consistent with the original co-holding result being driven by the mechanical link between formal employment and provident-fund participation.

Table 4: Co-holding and conditional high-cost-debt results

Specification	Formal-employment coefficient	p -value	Observations
Co-holding: primary model with INC_CV	0.0161	< 0.001	287,587
Co-holding: full model without INC_CV	0.0143	< 0.001	326,558
High-cost debt among PF savers	0.0003	0.9468	61,748

Note: WLS estimates with household-clustered standard errors. All specifications include head age, household size, state fixed effects, and education-group fixed effects. The first specification also includes income volatility. The final row conditions on provident-fund saving and changes the dependent variable from co-holding to high-cost debt.

5.3 Intention–Behaviour Gaps

Among households that indicated an intention to save in a provident fund, formal employment is associated with an 8.37-percentage-point lower failure-to-follow-through gap ($p < 0.001$). For gold, however, the coefficient is 1.74 percentage points and statistically insignificant ($p = 0.320$). The contrast is consistent with payroll deduction or institutional access explaining the provident-fund relationship, rather than formal employment creating a general advantage in voluntary saving follow-through.

Table 5: Formal employment and saving intention–behaviour gaps

Saving vehicle	Formal-employment coefficient	p -value	Observations
Provident fund	−0.0837	< 0.001	13,653
Gold	0.0174	0.3195	5,578

Note: WLS estimates among households that reported the relevant saving intention. Standard errors are clustered by household. Models include income volatility, head age, household size, state fixed effects, and education-group fixed effects. The gap is evaluated from the January–April to May–August transition.

5.4 Discussion

Three conclusions emerge. First, the permanent-income decomposition is consistent with consumption smoothing: the response to persistent income is substantially larger than the response to within-household temporary income movements. Second, income volatility does not have a statistically significant association with co-holding once the model is weighted and inference is clustered by household. Third, the formal-employment association in co-holding and provident-fund follow-through appears to be specific to the provident-fund institutional channel; it does not extend to gold saving.

These findings should be interpreted carefully. The models use observational survey data and cannot establish causality. In addition, the gap and co-holding constructs depend on the exact CMIE questionnaire definitions. The expenditure-unit check is reassuring but not a substitute for formal validation against the CMIE data dictionary.

H1 — Supported.

$\beta_P = 0.320 > \beta_T = 0.053$; Wald test rejects equality ($p < 0.001$).

H2 — Partially supported.

Formal employment is significant; income volatility is not.

H3 — Supported.

Formal employment reduces the PF gap by 8.37 pp ($p < 0.001$).

H4 — Not supported.

No significant formal-employment effect for gold saving ($p = 0.320$).

6 Conclusion

6.1 Summary and Answers to the Research Questions

Using a cleaned CMIE household panel for 2025, this study finds a positive association between income and expenditure and evidence that consumption responds more strongly to permanent than transitory income. It documents a non-trivial co-holding rate of high-cost debt and illiquid savings, but finds no robust evidence that income volatility predicts this pattern. Formal employment is associated with raw co-holding and a smaller provident-fund gap; however, this association disappears when controlling for existing provident-fund savings, and no such association exists for gold. The most plausible interpretation is therefore institutional rather than a broad behavioural advantage of formal employment.

6.2 Contributions

The paper makes three contributions. First, it assembles multiple CMIE modules into a unified household panel while preserving non-response and family-shift information. Second, it combines monthly consumption-smoothing tests with wave-level portfolio-choice analysis. Third, it applies targeted checks that distinguish a mechanical payroll-saving explanation from a general formal-employment effect.

6.3 Limitations

The analysis has four principal limitations. First, CMIE survey variables may not perfectly measure the economic concepts of interest, particularly expenditure frequency and high-cost debt. Second, the sample is unbalanced and missingness may be systematically related to household characteristics. Third, the models are reduced-form associations and omit potentially relevant time-varying shocks. Fourth, the study covers one calendar year, limiting long-run life-cycle inference.

6.4 Future Research

Future work should validate all monetary variables against the CMIE data dictionary, incorporate additional years to study persistence and transitions, estimate fixed-effects or event-study specifications around income shocks, and distinguish debt sources by interest rate, collateral, and repayment terms. Linking household outcomes to local financial-access

measures would also help identify whether geography and institutional supply shape the observed saving and debt choices.

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